Exercise 1 : SQL Fundamentals

Questions

1.) SELECT Statement

Write a SQL query to retrieve all columns from the employees table.

```
SELECT *
FROM employees;
```

Result

id	first_name	last_name	department	salary	hire_date	city
1	John	Doe	IT	55 000	2018-09-15	New York
2	Jane	Smith	HR	48 000	2019-07-20	Chicago
3	Mike	Johnson	Finance	60 000	2017-09-30	Los Angeles
4	Sarah	Brown	IT	58 000	2021-03-25	New York
5	David	White	Marketing	52 000	2016-04-10	San Francisco
6	Emily	Davis	IT	62 000	2015-02-14	Chicago
7	Robert	Wilson	Finance	59 000	2019-10-01	Houston
8	Jessica	Moore	HR	51 000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53 000	2022-06-01	Chicago
10	Laura	Hall	IT	50 000	2020-08-10	San Francisco

2.) SQL Query

```
SELECT DISTINCT department
FROM employees
```

Result

department
IT
HR
Finance
Marketing

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rketing

3.) ORDER BY
SQL Query

```
SELECT first_name, last_name, salary
FROM employees
ORDER BY salary DESC;
```

Result

first_name	last_name	salary
Emily	Davis	62 000
Mike	Johnson	60 000
Robert	Wilson	59 000
John	Doe	55 000
Sarah	Brown	53 000
Daniel	Clark	53 000
David	White	52 000
Jessica	Moore	51 000
Laura	Hall	50 000
Jane	Smith	48 000

4.) LIMIT
SQL Query

```
SELECT *
FROM employees
ORDER BY salary DESC
LIMIT 5;
```

Result

id	first_name	last_name	department	salary	hire_date	city
6	Emily	Davis	IT	62000	2015-02-17	Chicago
3	Mike	Johnson	Finance	60 000	2017-09-30	Los Angeles
7	Robert	Wilson	Finance	59 000	2019-10-01	Houston
1	John	Doe	IT	55000	2018-06-15	New York
4	Sarah	Brown	IT	53 000	2021-03-25	New York

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11.	Sarah	Brown	IT	63 000	2021-05-20 New York

5.) WHERE
SQL Query

```
SELECT *
FROM employees
WHERE department = 'IT';
```

Result

first_name	last_name	salary	city
John	Doe	55000	New York
Sarah	Brown	58000	New York
Emily	Davis	62000	Chicago
Laura	Hall	50000	San Francisco

6.) AND
SQL Query

```
SELECT *
FROM employees
WHERE department = 'Finance'
AND salary > 58000;
```

Result

first_name	last_name	salary
Mike	Johnson	60000
Robert	Wilson	59000

7.) OR
SQL Query

```
SELECT *
FROM employees
WHERE department = 'HR'
OR department = 'Marketing';
```

Result

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WHERE department = 'HR'
OR department = 'Marketing';

Result

first_name	last_name	department
Jane	Smith	HR
Jessica	Moore	HR
David	White	Marketing
Daniel	Clark	Marketing

8.) NOT
SQL Query

```
SELECT *
FROM employees
WHERE department <> 'IT'
```

Result

first_name	lastname	department
Jane	Smith	HR
Mike	Johnson	Finance
David	White	Marketing
Robert	Wilson	Finance
Jessica	Moore	HR
Daniel	Clerk	Marketing

9.) IN
SQL Query

```
SELECT *
FROM employees
WHERE department IN ('HR', 'IT', 'Finance')
```

first_name	Last_name	department
John	Doe	IT
Jane	Smith	HR
Mike	Johnson	Finance

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WHERE department IN ('HR', 'IT', 'Finance')

first_name	last_name	department
John	Doe	IT
Jane	Smith	HR
Mike	Johnson	Finance
Sarah	Brown	IT
Emi	Davis	IT
Robert	Wilson	Finance
Jessica	Moore	HR
Laura	Hall	IT

10.) Combining Conditions
SQL Query

```
SELECT *
FROM employees
WHERE department = 'IT'
    AND salary > 5000
    AND city = 'New York';
```

Result

first_name	last_name	salary	city
John	Doe	5000	New York
Sarah	Brown	53000	New York

11.) WHERE + AND + ORDER BY
SQL Query

```
SELECT first_name,
       last_name,
       salary
FROM employees
WHERE department IN ('Finance', 'Marketing')
    AND salary > 52 000
ORDER BY salary DESC;
```

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DER BY salary DESC;

Result

first_name	last_name	salary
Mike	Johnson	60 000
Robert	Wilson	59 000
Daniel	Clark	63 000

12.) Distinct + Where + IN
SQL Query

```
SELECT DISTINCT city
FROM employees
WHERE department NOT IN ('IT','HR');
```

Result

City
Los Angeles
San Francisco
Houston
Chicago

13.) NOT + AND + ORDER BY
SQL Query

```
SELECT *
FROM employees
WHERE department <> 'Finance'
    AND salary > 50000
ORDER BY hire_date ASC;
```

Result

```
WHERE department = 'Finance'
AND
ORDER BY hire_date ASC;
```

Result

first_name	last_name	salary	hire_date
Emily	Davis	62000	2015-02-14
David	White	52000	2016-04-10
John	Doe	55000	2018-06-15
Jessica	Moore	51000	2018-05-22
Sarah	Brown	53000	2021-03-25
Daniel	Clark	53000	2022-06-01

14) OR + IN + LIMIT

SQL Query

```
SELECT *
FROM employees
WHERE city IN ('Chicago', 'Los Angeles')
  AND department IN ('IT', 'Marketing')
LIMIT 3;
```

Result

first_name	last_name	department	city
Emily	Davis	IT	Chicago
Daniel	Clark	Marketing	Chicago

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